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Abstract

This report illustrates the development of a step ladder that provides passengers and flight attendants inside an aircraft cabin with assistance reaching for carry-on baggage in the overhead compartments. By following design constraints, this project hopes to improve flight attendant and passenger safety.

The design revolves around an iterative procedure with key constraints, starting with reviewing existing literature, ranging from reports to drawings. In doing so, an increasingly adequate conceptual understanding provided more specific design considerations to adhere to and potential solutions to the issue. The most important design considerations include but are not limited to material selection, weight, load capacity, portability, and cost. Materials consist of polyurethane-coated aluminum for the frame and softened polyethylene/silicone for the grips. The result was a product that weighed roughly 10 lbs, was light enough to use, and complied with FAA standards. The load capacity is 250 lbs to fulfill safety regulations, while portability was upheld through weight and ergonomics. Furthermore, the manufacturing cost ensures economic viability. After various attempts, the final product was selected using a design matrix, resulting in a scissor-like mechanism that allows for compact storage and height customization.

To model the final product, a physical prototype was not possible. However, using CATIA, a CAD model was developed to show the assembled version alongside all of the individual parts, with orthographic drawings consisting of appropriate views and dimensions to ensure feasibility. This provides a proof of concept, which ensures the design is not just theoretical but practical.

The resulting product is a success, demonstrating the overall goal of improving flight safety. All parameters were fulfilled, and all necessary drawings and documentation were proof of project feasibility, only possible with the collective contribution of the group and guidance from our TA. Although there could be improvements, the end product provides additional inspiration for further design and other potential products to improve aviation.

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Introduction

Step ladders and step stools play key roles throughout various situations, allowing a safe and stable way to support individuals unable to reach areas high above the ground, such as

kitchens, bathrooms, basements, and attics. However, in the context of this course, the most common hard-to-reach areas would be within the aircraft cabin itself. This is an issue for passengers and crew because of the unusual elevation of overhead compartments. Here, accessibility is a key consideration, given that not everyone is of equal height or physical strength, making it difficult for some to stow away their carry-on luggage. Typical step stools and step ladders suffer various issues in the context of aircraft cabins. One of the key issues is a lack of storage, as many step stools and step ladders do not compress to a size for storing under seats. Thus, it violates FAA regulations. Furthermore, most step stools that are small enough can not provide for different heights, making it unideal given the diverse population that uses airlines.

The key motivation for this project is the need for efficient and user-friendly designs to address the limited accessibility issue while still complying with the various FAA standards. In light of this, various design challenges must be addressed, including weight, size, and portability restrictions. This is particularly important in an aircraft cabin, which is narrow and has many people walking throughout. By improving the accessibility of overhead compartments, the potential for muscle strain for flight attendants is less of an issue, and slow boarding due to struggle and risk of injury for passengers is not as significant. Thus, such a product would be useful in the airline industry, particularly for flights with a large volume of passengers.

The scope of this project extends beyond just a conceptual idea. Instead, it must be a feasible and practical one. The design process and the rest of the report follow a step-by-step process of researching, brainstorming, conceptual design, detail design, analysis, reiteration if necessary, and evaluation. This ensures a well-thought-out final product that meets the various criteria. The report consists of various sections:

1. Literature review: reports on existing designs, industry standards, flaws, and potential engineering concepts required for understanding the rest of the report.
2. Design statement: provides a concise idea of the constraints and variables.
3. Design process and conceptual design: detailed description of potential ideas and rationale behind the chosen design.
4. Writing and documentation: provides the necessary documentation and detailed explanation of the chosen design, including figures, tables, and calculations.
5. Conclusion: Summarizes reports, including all findings.

6. Future design considerations: Describes various improvements that could be made to the current design through current methods, alongside a statement of contribution.

Literature Review & Existing Designs

The challenge of placing carry-on luggage into aircraft overhead compartments has long been recognized, specifically for elderly passengers, slightly short people, and flight attendants who encounter physical strain due to repeated lifting. This section explores relevant standards, engineering principles and existing products to identify the design gaps and aid the development of a compact and portable aircraft cabin step ladder.

Whilst designing equipment for aircraft, it is necessary to adhere to the safety and design regulations. The Federal Aviation Administration (FAA) and Occupational Safety and Health Administration (OSHA) provide the required guidelines that shape the development of aviation-related support tools.

According to FAA regulations [1], all portable equipment used in cabins must be fire-resistant, non-toxic, and safe to operate under light aircraft turbulence. OSHA 1910.25 outlines safety provisions for ladders and step tools, including uniform step spacing (<8 inches) and parallel and level steps in the usage position.

Aircraft cabins cause specific challenges on step ladder designs due to their limited aisle width (minimum 0.46m), weight limitations, and storage constraints. The Airbus A350, selected as a reference for this design, provides a storage volume of around 14" × 15" × 8.8" under the flight attendant seating, which gives the requirement of designing a step ladder that is able to fold and fit within this confined space.

Additionally, due to the growing aging population and passengers with physical disabilities, providing an accessible solution to storing luggage in overhead bins has become essential.

	 DELTA ONE® SUITES	 DELTA PREMIUM SELECT	 DELTA COMFORT+	 MAIN CABIN
ACCOMMODATION (306 SEATS)	32 Seats	48 Seats	36 seats	190 Seats
SEAT WIDTH/PITCH	20.46 in / 76-77 in layflat bed	18.5 in/38 in 47 cm/97 cm	17.4 in/34 in 44 cm/86 cm	17.4 in/31-32 in 44 cm/79- 81cm
CHILD SEAT FIT GUIDE Aircraft seat width min-max	20.46 in 52 cm	18.5 in 47 cm	17.37 in 44.1 cm	16.19 in 41.1 cm
UNDERSEAT DIMENSIONS (depth x width x height)	No underseat storage space	14 in x 18 in x 5.5 in	14 in x 15 in x 8.8 in	14 in x 15 in x 8.8 in
AMENITIES				

Fig 1. Dimensions of the interior for the Airbus A350 [8]

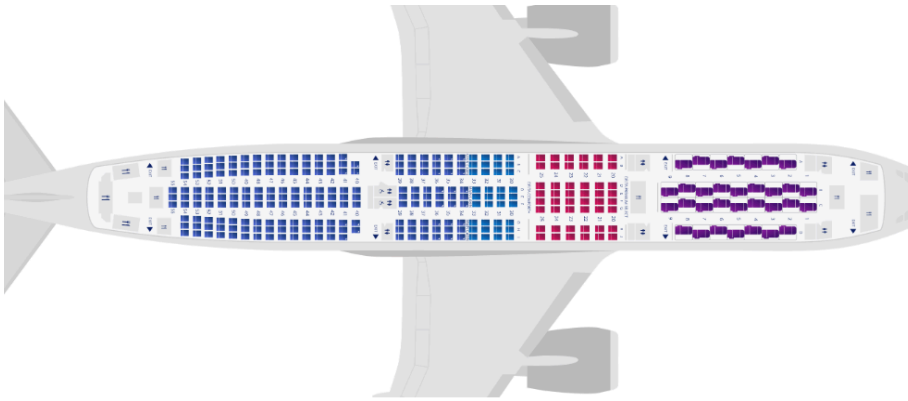


Fig 2. Layout of the interior for the Airbus A350 [8]

Several step ladders exist in the market; however, none are suitable to be used in aircraft as they lack safety features and are too bulky. Some of the existing step ladders are as follows:

1. Generic folding step ladders. These designs are common and inexpensive, and whilst these designs could vary slightly, they all fail to meet the aviation standards due to incompatible dimensions or lack of locking mechanisms.



Figure 3. An existing foldable ladder [3]



Figure 4. An example of a telescoping ladder [4]

2. Telescoping step ladders. Some of these models use telescoping mechanisms to collapse into compact forms, which inspires modular height-adjustable features. However, the locking reliability is a concern.

From the reviewed designs, a consistent issue lies in the lack of lightweight, cabin-compliant solutions for passengers rather than maintenance staff. Therefore, a step ladder needs to be developed to be compact, foldable and lightweight in order to be able to be used in aircraft. It must also be made easy to use and support a weight of up to 250lbs.

Design Statement

Based on the understanding of prior products, the design statement will provide a more specific guideline to modify the previous design to suit the project needs. Regarding the design objectives:

- Flight attendant safety: prevent physical strain and the potential for severe injuries through a user-friendly, safe, and ergonomic method to reach carry-on luggage in overhead compartments
- Passenger assistance: allow passengers additional autonomy, providing them with a means of reaching overhead compartments regardless of height or strength
- Convenience: reduce boarding and deboarding times by quickly enabling passengers to get their own luggages

- Must also be lightweight for the sake of usability
- Must be foldable to fit under the seats of passengers or flight crew
- Must have a quick-to-use design through ergonomics
- Safety and stability: Must have anti-slip surfaces through good material selection for the bottom and the surface that people will step on, decreasing the chances of slipping or tripping during use
 - Must include a redundant locking mechanism to prevent collapse
- Comply with FAA and EASA regulations
- Longevity: must be durable for repeated use in rapidly changing conditions of varying temperatures, humidity, pressure, etc.
- Environmental concerns: Use materials that are capable of lasting a while without the need for replacement
 - Use recyclable materials to ensure minimal carbon footprint
 - Should be low in cost to allow for mass production and distribution
- Visual appeal: Should have neutral colours and complement aircraft

To appeal to the design objectives, it is key to follow specific numerical design constraints, which include size, weight, height, load, code of conduct, deployment, and cost:

1. Storage:
 - a. For the sake of consistency, the Airbus A350, a commercial aircraft often used for long-haul flights, will be used as the basis of the design.
 - b. The space under the flight attendant seats is 14 inches by 15 inches by 8.8 inches. Thus, the folded max dimensions are such.
 - c. The cabin walkway is 0.46m wide at the smallest. Thus, the unfolded dimensions should not exceed 0.46m wide, and ideally should be smaller to allow for less traffic while boarding and deboarding the aircraft.
2. Weight: for the sake of usability, the product must not exceed 12lbs, allowing everyone access regardless of strength and to maintain weight under limits
3. Height: provide a height boost of up to 6 inches in order to allow those below the height threshold to reach for the overhead compartment
4. Load: must be able to support 200 lbs of weight to maintain FAA regulations
5. Convenience: must be deployed within 8s (stowed away and in use)

6. FAA and OSHA code of conduct:
 - a. Must be non-toxic, spark-resistant, and fire-resistant [1]
 - b. Must maintain position during light aircraft movement [1]
 - c. OSHA 1910.25: relevant safety regulations
 - i. A uniform step spacing of less than 8 inches [1], [2]
 - ii. Steps shall be parallel and level when the ladder or stool is in position for use [2]
 - iii. Rails and protective cages shall be present if beyond 24 inches in height [2]
 - d. ANSI ASC A14.2: Design & Testing Step Ladders
 - i. Load capacity of at least 250 lbs [1], [2]
 - ii. Anti-slip and handles [1], [2]
7. General use: ensure minimum pinch points and an intuitive design to prevent misuse

With the functional requirements, constraints, and design variables noted with distinct numerical values when possible, brainstorming potential solutions can begin.

Design Process and Conceptual Design

Several Design ideas came into consideration while trying to design this step ladder.

Drawing of the first design was this:

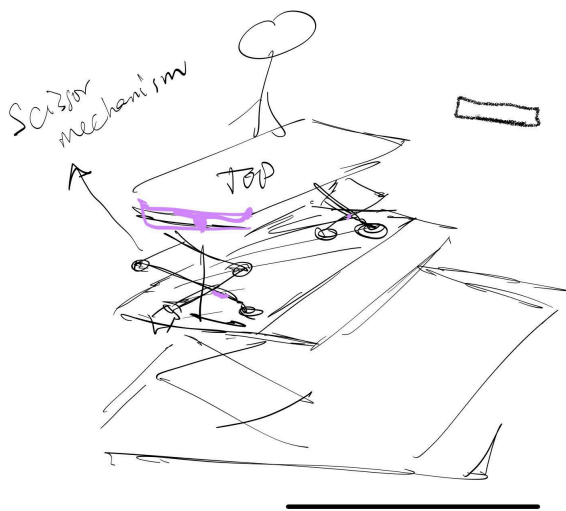


Figure 5: Initial sketch for the first design idea

This design was promising; however, it checked the boxes for our design process. The boxes were the step ladder being able to collapse and fit under the aisle seat chairs, being lightweight. The scissor mechanism was a great idea, but there was a challenge as the scissor mechanism would need pneumatic or hydraulic power to expand and contract the steps. Storing this step ladder was difficult as it would have wheels. This was a no, as compressed gas is seen as a safety hazard and also increased the overall weight. Maintenance costs and price will increase the overall expenses for Delta airline. The second design resulted in this:

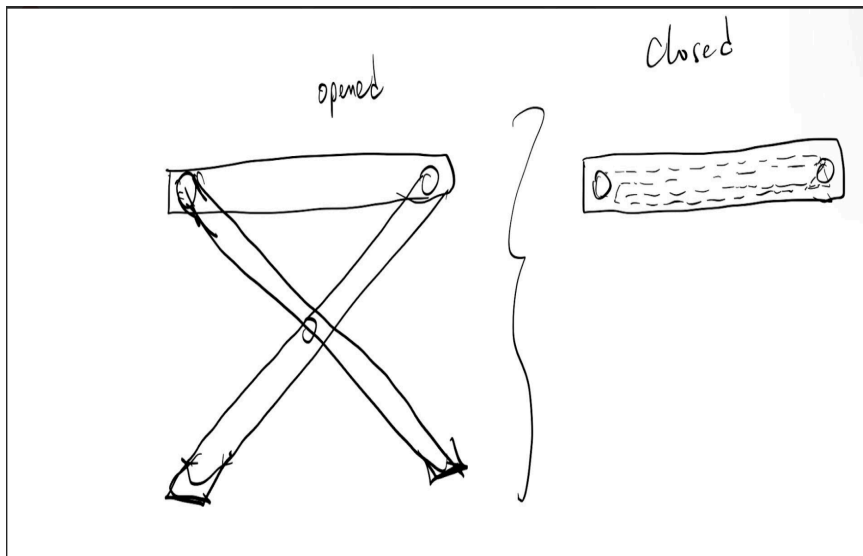


Figure 6: First reiteration of the initial design

This design was a basic and simple design which was small and efficient to use during boarding time on Delta Airlines. Our group designed it so that the step would be made from hard plastic and the legs from hollow aluminum; this kept the weight to 10 lbs, which was great. But the problem with this design was that the carry capacity would be low, being under 130 lbs. This could be an issue as it could bring lawsuits for Delta Airlines or any other airline company that uses it. This stool worked by using hinges underneath it and a locking mechanism in the legs to lock it in place.

Detailed Design & Analysis

After several brainstorming sessions, my partners and I came up with an idea that we believed solved the problems from our previous designs. This step ladder consisted of one step; it was

light enough to be easily moved while being able to support up to 250lbs. These are the first non-dimensioned sketches of our step ladder.

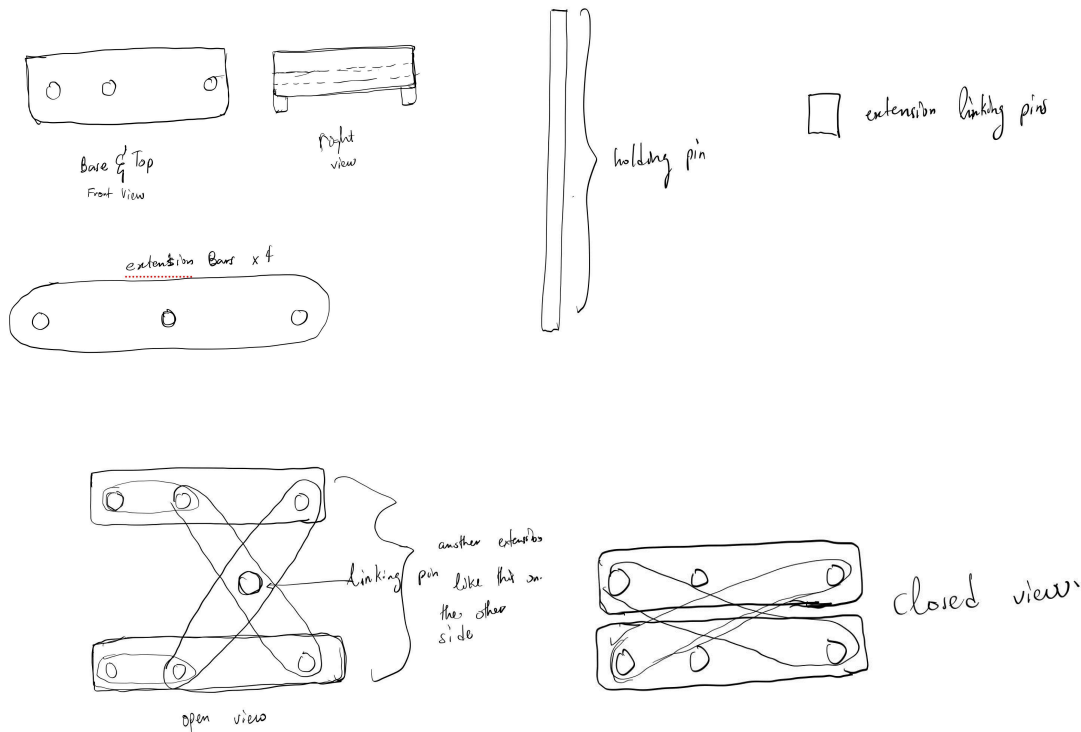


Figure 7: Final rough sketches of the final design

This ladder employed the use of a scissors mechanism; however, it used only one extension and not several as the first design and used a rigged locking mechanism to lock the extension bars in place. The scissors mechanism did support the theory of triangles being the strongest shape to hold weight, as when opened, the shape formed two triangles.

These are the pictures of each part designed in Catia V5. Designing the parts of the ladders was quite easy; however, assembling proved quite difficult despite the measurements having been taken into account. There were potential design changes to make it possible to assemble. The pin locking mechanism was not shown on CAD, but the locking mechanism we are using will be implemented as it is a solid mechanism that will work and help in efficiency when opening and closing the step ladder.

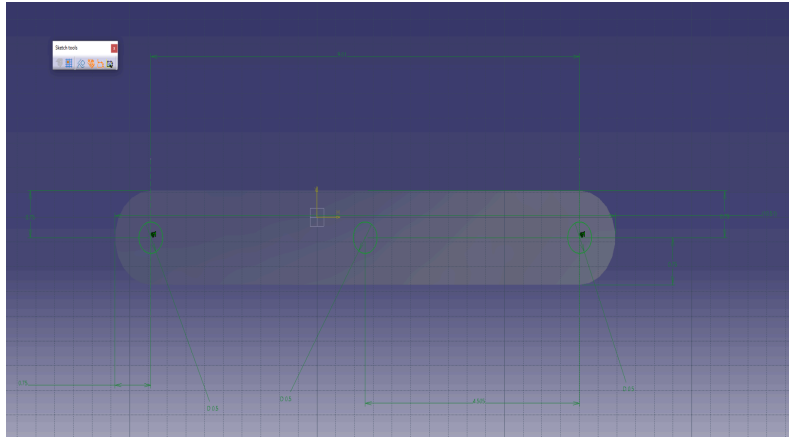


Figure 8. Extension bar

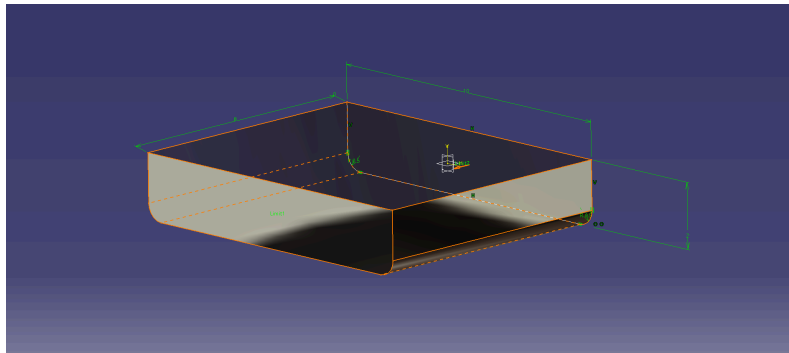


Figure 9. Top and base

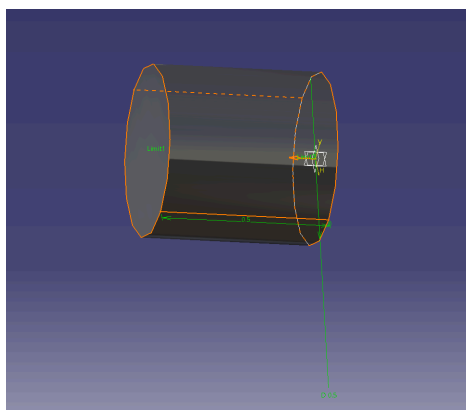


Figure 10. Extension bar joining pin

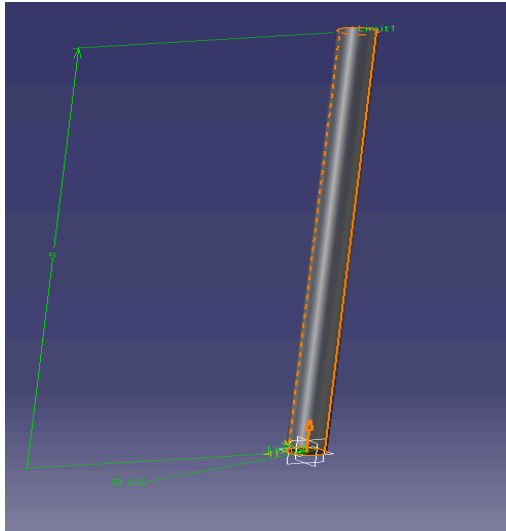


Figure 10. Long pin that attaches the extension bar and the top

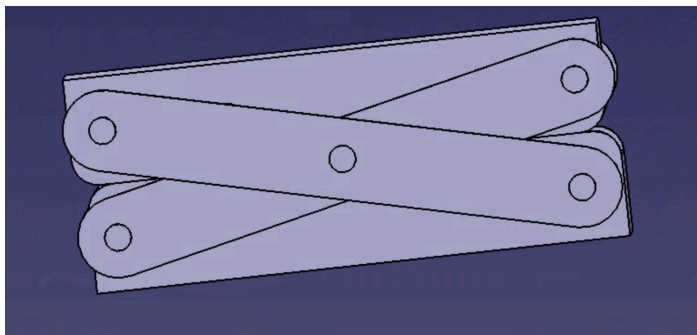


Figure 11. Assembled side view

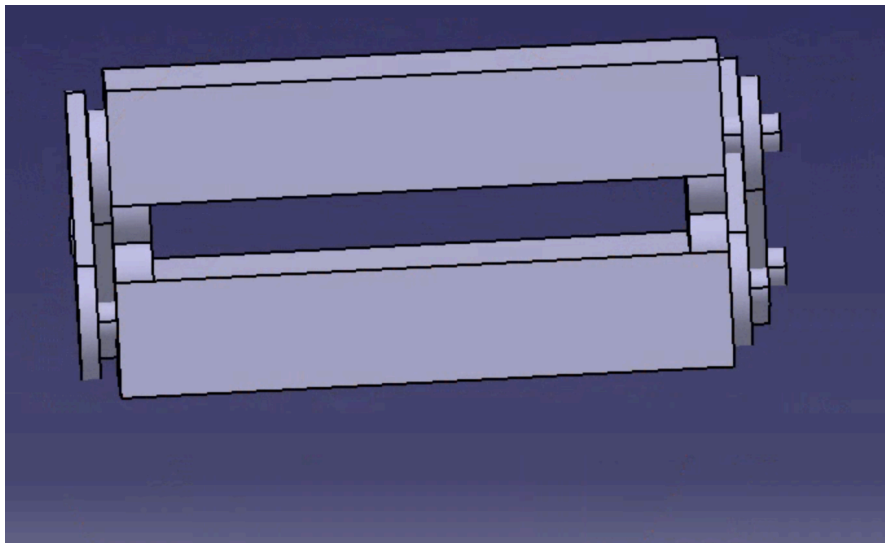


Figure 12. Assembled front view with the assembly slightly expanded

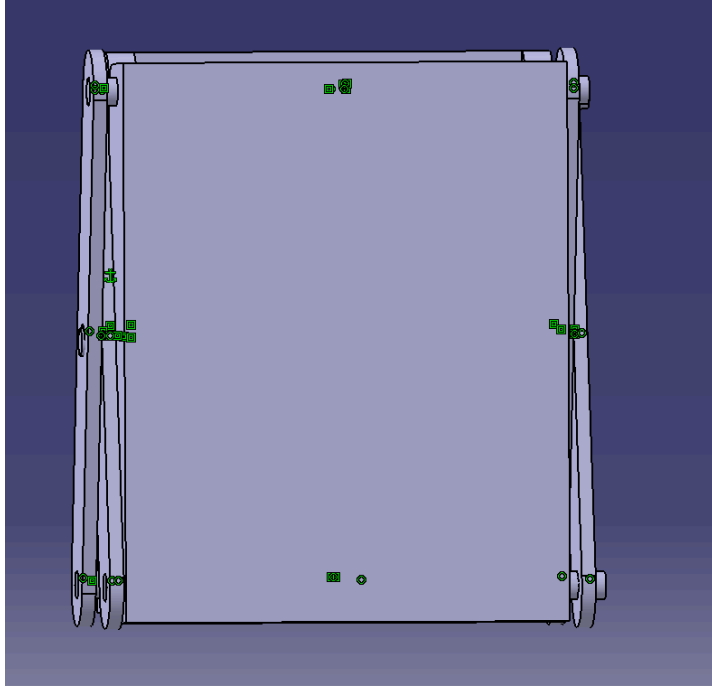


Figure 13. Assembled top view

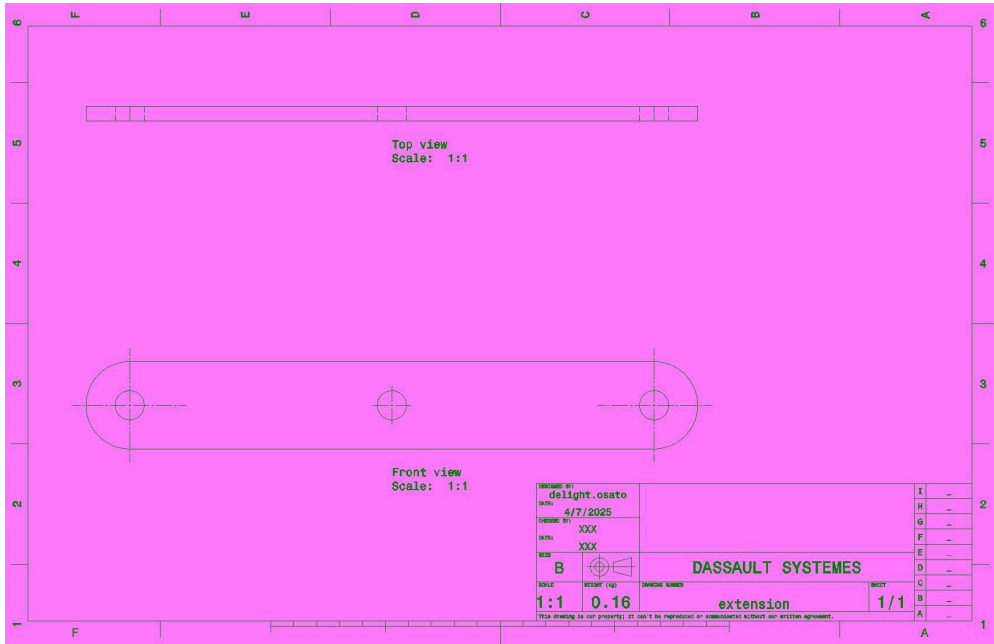


Figure 14. Orthographic drawings of the extension bar

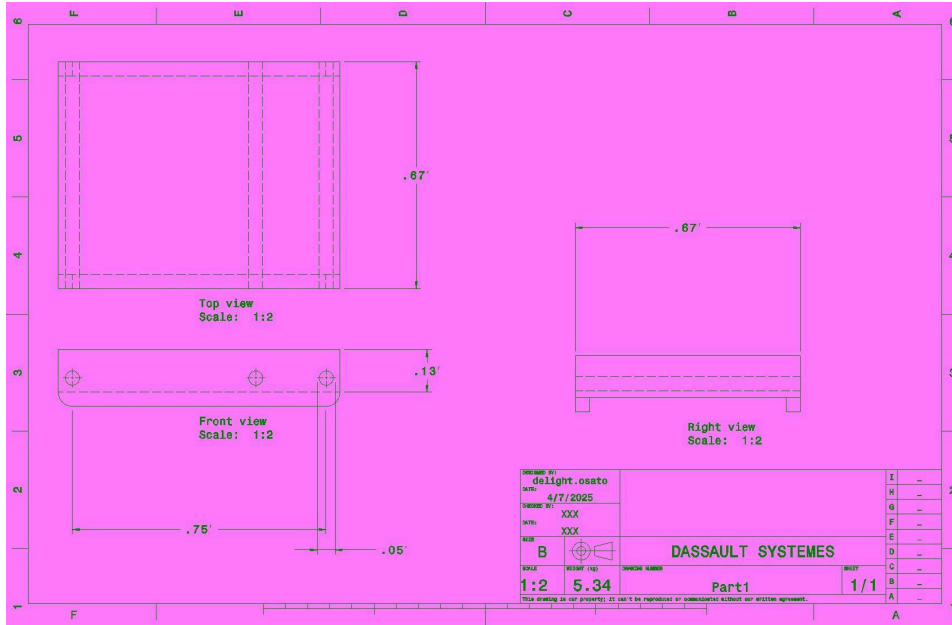


Figure 14. Orthographic drawings of the base

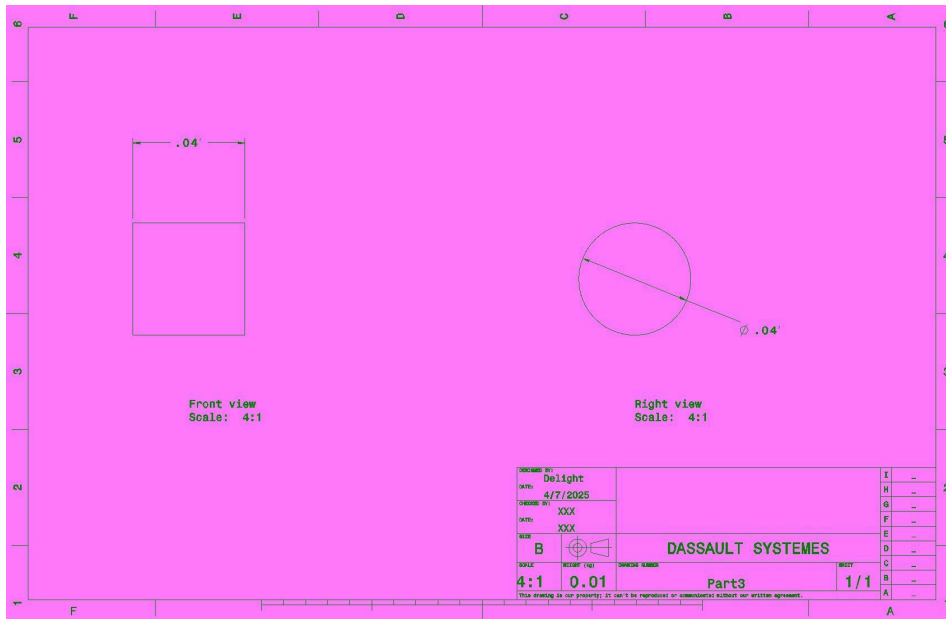


Figure 15. Orthographic drawings of the extension bar joining the bin and top

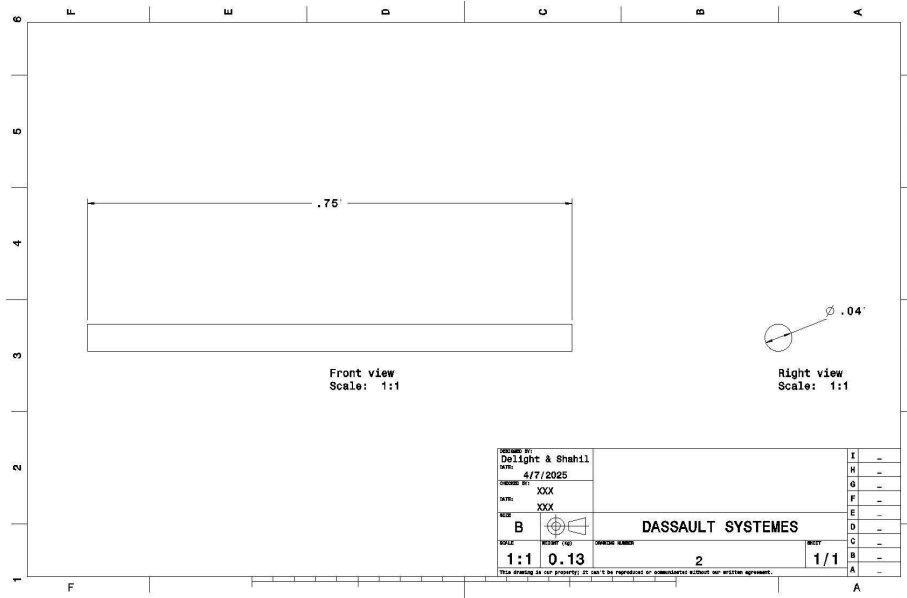


Figure 16. Orthographic drawings of the pin

Table 1. Materials List Including Cost per Gram

Item name	Material	Price (according to various resources)
Long Pin & short pin	Titanium	\$6.61 / 1 gram
Extension piece	Aluminum	\$0.17-\$2.37 / lbs
Base piece	Aluminum	\$0.17-\$2.37 / lbs
Side piece	Plastic	\$0.77 / lbs

These costs would be multiplied by the overall weight of the items. Manufacturing and labour costs would also have to be considered to find the total cost of the step ladder

Conclusions

In conclusion, the final design for the aircraft cabin step ladder significantly solves the issue that mainly cabin crew or passengers might face during the boarding or exiting of an aircraft. Throughout the design process, accessibility was a main consideration to ensure inclusivity for all users. Through research, review of existing designs and multiple design ideas,

a functional and realistic final design solution was created through CATIA along with multiview drawings to clearly communicate the many parts and functions of the step ladder.

Future Design Considerations

Looking ahead, many design considerations must be considered to ensure long-term sustainability and functionality.

Deciding which materials to use in the product is one of the most important parts of product improvement. For the design of the step ladder, materials such as aluminum, titanium, and plastic were selected as they are lightweight, strong materials, and are resistant to corrosion/rust, which makes them ideal for the demanding conditions of aircraft environments. However, future design ideas may explore the use of even lighter materials to reduce the overall weight of the step ladder but should still be able to withstand heavy weights.

Secondly, improving the folding mechanism to allow for smoother operation and more compact storage would also benefit usability, especially in more compact aircraft environments. Designs that are easy to open/close and efficient stowage could significantly improve user experience and operational convenience.

Lastly, design improvements to the exterior of the step ladder, such as environmental resistance, could also be taken into consideration. Due to the varying and harsh conditions inside aircraft, materials must be able to withstand fluctuating temperatures, changing pressure levels, high humidity, etc. These environmental factors can lead to corrosion or general degradation of the product over time. Protective surface treatments, such as thin polymer coatings or weather-resistant finishes, could improve the ladder's long-term durability and performance. These coatings can act as barriers against moisture and varying temperatures, helping to extend the product's lifespan while maintaining its safety and reliability in diverse operating conditions.

References

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